PROJECT BRIEF · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 6: Project: Live Leaderboard

Module 2 | 50 minutes | Project: sorting app | Grades 6–8 · ages 12–14

Name: _____ Class: _____ Date: _____

Your mission
Build a leaderboard: collect any number of scores into a list, sort them highest-first with your sorting procedure, announce the top three, and handle edge cases like fewer than three players. Test it with planned test sets.

Checklist

- [ ] Collects scores
- [ ] Sorts descending
- [ ] Top three
- [ ] Edge cases
- [ ] Procedures
- [ ] Test sets

Milestones

1. Collection — done on: _____
2. Sorting — done on: _____
3. Reporting — done on: _____
4. Edge cases — done on: _____
5. Tested and submitted — done on: _____

Plan it

Sketch your stage and write what each sprite will do.

How it will be marked

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Sorting	Not sorted	Partly sorted	Correctly sorted descending	Efficient (early stop) and correct
Input & output	Fixed data	Collects scores	Collects and reports top three	Clear, friendly announcements
Edge cases	Breaks	One handled	Small groups handled	Small groups and ties handled
Procedures	None	swap only	swap and sort procedures	Clean modular design
Testing	None	One run	Four planned test sets	Test sets with expected vs actual

Want more?

- Player names in a parallel list
- Early-stop bubble sort
- Bar chart of scores

Two stars and a wish

Ask a friend to test your project.

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